their source rather than the Earth and when the Earth is feeling them rather than you, they produce the ocean tides. See Box 2.5.

In deducing his principle of equivalence, Einstein ignored tidal gravitational forces; he pretended they do not exist. (Recall the essence of his argument: As you fall freely, you "will not feel your own weight" and "it will seem to you, in all respects, as though gravity has disappeared from your vicinity.") Einstein justified ignoring tidal forces by imagining that you (and your reference frame) are very small. For example, if you are the size of an ant or smaller, then your body parts will all be very close to each other, the direction and strength of gravity's pull will therefore be very nearly the same on the outer parts of your body as at its center, and the *difference* in gravity between your outer parts and your center, which causes the tidal stretch and squeeze, will be extremely small. On the other hand, if you are a 5000-kilometer-tall giant, then the direction and strength of the Earth's gravitational pull will differ greatly between the outer parts of your body and its center; and correspondingly, as you fall, you will experience a huge tidal stretch and squeeze.

This reasoning convinced Einstein that, in a sufficiently small, freely falling reference frame (a frame very small compared to the distance over which gravity's pull changes), one should not be able to detect any influences of tidal gravity whatsoever; that is, small, freely falling reference frames in our gravity-endowed Universe are equivalent to inertial frames in a universe without gravity. But not so for large frames. And the tidal forces felt in large frames seemed to Einstein, in 1911, to be a key to the ultimate nature of gravity.

Box 2.5

Ocean Tides Produced by Tidal Forces